

Extended Results

Results too voluminous for the main manuscript. All are regenerated by code/reproduce_all.py.

E1. Noise experiment: full distributional summary

Per-realization records: 400 rows (100 realizations at each of 4 SNR levels), master seed 20260826. The complete record is in data/signal_corpus/S6_noise/per_realization.csv. Quartiles are given so that the distribution can be inspected directly rather than summarized only by mean and standard deviation.

	snr_db	0.0	10.0	20.0	30.0
eta_total	count	100	100	100	100
	mean	0.95811	0.98812	0.99136	0.99171
	std	0.016028	0.0028644	0.0010349	0.00029038
	min	0.90456	0.98072	0.98861	0.99104
	25%	0.95086	0.98626	0.9907	0.99146
	50%	0.95943	0.98808	0.99141	0.99176
	75%	0.96818	0.98977	0.99209	0.99192
	max	0.98681	0.99471	0.99381	0.99239
eta_ac	count	100	100	100	100
	mean	0.65906	0.83807	0.87459	0.87874
	std	0.11795	0.037037	0.014212	0.0040847
	min	0.31564	0.75149	0.83802	0.86909
	25%	0.59188	0.81118	0.86533	0.8757
	50%	0.68579	0.84072	0.87506	0.87887
	75%	0.74468	0.86275	0.88512	0.8818

	snr_db	0.0	10.0	20.0	30.0
eps_l2_vs_clean	max	0.88813	0.91662	0.90673	0.88914
	count	100	100	100	100
	mean	0.19222	0.10473	0.092469	0.091
	std	0.031915	0.0051464	0.0006601	7.3153e-05
	min	0.12017	0.093721	0.091248	0.090884
	25%	0.17282	0.1012	0.092012	0.090951
	50%	0.18877	0.10398	0.092379	0.090986
	75%	0.21513	0.10787	0.092885	0.091039
	max	0.27271	0.11784	0.094892	0.091383
	count	100	100	100	100
eps_l2_ac_vs_clean	mean	0.71508	0.39575	0.35353	0.34848
	std	0.12219	0.018556	0.0024425	0.0002629
	min	0.45515	0.3585	0.34933	0.3481
	25%	0.62652	0.38239	0.35184	0.3483
	50%	0.70915	0.39355	0.35312	0.34845
	75%	0.80944	0.4065	0.35514	0.34861
	max	1.0423	0.44101	0.36318	0.34988
	count	100	100	100	100

E2. Complete retention curves

All 324 retention measurements across corpus members, strategies and bases. These are the measurements against which the identity $\varepsilon_{L2} = \sqrt{(1-\eta)}$ was tested.

n_co eff	coeff_ratio	eta_total	eta_ac	eps_l2	rmse	snr_db	max_imag	strategy	basis	id
1	0.0384615	0.931834	0	0.261087	66.6391	11.6643	0	lowpass	total	S0
3	0.115385	0.935683	0.0564705	0.253608	64.7302	11.9167	0	lowpass	total	S0
5	0.192308	0.96859	0.539216	0.177228	45.2353	15.0293	0	lowpass	total	S0
7	0.269231	0.976063	0.648847	0.154715	39.4892	16.2093	0	lowpass	total	S0
9	0.346154	0.990637	0.862652	0.0967602	24.6968	20.2861	0	lowpass	total	S0
11	0.423077	0.991747	0.87893	0.0908455	23.1872	20.8339	0	lowpass	total	S0
13	0.5	0.993773	0.908652	0.0789105	20.1409	22.0573	0	lowpass	total	S0
15	0.576923	0.994864	0.92466	0.0716634	18.2912	22.894	0	lowpass	total	S0
17	0.653846	0.995362	0.931966	0.0681001	17.3817	23.337	0	lowpass	total	S0
19	0.730769	0.996398	0.947151	0.0600208	15.3195	24.434	0	lowpass	total	S0
21	0.807692	0.997695	0.966192	0.0480059	12.2529	26.3741	0	lowpass	total	S0
23	0.884615	0.998128	0.972544	0.043262	11.0421	27.2779	0	lowpass	total	S0
25	0.961538	0.998881	0.983583	0.033453	8.53846	29.5113	0	lowpass	total	S0
26	1	1	1	2.50703e-15	6.39889e-13	292.017	0	lowpass	total	S0
2	0.0769231	0.00384939	0.0564705	0.971354	64.7302	0.252445	0	lowpass	ac	S0
4	0.153846	0.0367564	0.539216	0.67881	45.2353	3.36503	0	lowpass	ac	S0
6	0.230769	0.0442295	0.648847	0.592582	39.4892	4.54503	0	lowpass	ac	S0
8	0.307692	0.0588038	0.862652	0.370605	24.6968	8.62177	0	lowpass	ac	S0
10	0.384615	0.0599134	0.87893	0.347951	23.1872	9.16963	0	lowpass	ac	S0
12	0.461538	0.0619395	0.908652	0.302238	20.1409	10.393	0	lowpass	ac	S0

n_co eff	coeff_ratio	eta_total	eta_ac	eps_l2	rmse	snr_db	max_imag	strategy	basis	id
14	0.538462	0.0630307	0.92466	0.274481	18.2912	11.2297	0	lowpass	ac	S0
16	0.615385	0.0635287	0.931966	0.260833	17.3817	11.6727	0	lowpass	ac	S0
18	0.692308	0.0645639	0.947151	0.229888	15.3195	12.7697	0	lowpass	ac	S0
20	0.769231	0.0658618	0.966192	0.18387	12.2529	14.7098	0	lowpass	ac	S0
22	0.846154	0.0662947	0.972544	0.1657	11.0421	15.6136	0	lowpass	ac	S0
24	0.923077	0.0670472	0.983583	0.12813	8.53846	17.847	0	lowpass	ac	S0
25	0.961538	0.0681663	1	9.47025e-15	6.31089e-13	280.473	0	lowpass	ac	S0
1	0.0384615	0.931834	0	0.261087	66.6391	11.6643	0	magnitude	total	S0
3	0.115385	0.964741	0.482746	0.187775	47.9271	14.5273	0	magnitude	total	S0
5	0.192308	0.979315	0.696551	0.143823	36.709	16.8434	0	magnitude	total	S0
7	0.269231	0.986788	0.806181	0.114943	29.3378	18.7903	0	magnitude	total	S0
9	0.346154	0.990637	0.862652	0.0967602	24.6968	20.2861	0	magnitude	total	S0
11	0.423077	0.992664	0.892374	0.0856533	21.8619	21.3451	0	magnitude	total	S0
13	0.5	0.993961	0.911414	0.0777082	19.834	22.1907	0	magnitude	total	S0
14	0.538462	0.995081	0.927831	0.0701389	17.9021	23.0808	0	magnitude	total	S0
16	0.615385	0.99619	0.94411	0.0617239	15.7543	24.1909	0	magnitude	total	S0
18	0.692308	0.997281	0.960118	0.0521404	13.3082	25.6565	0	magnitude	total	S0
20	0.769231	0.998317	0.975303	0.0410304	10.4725	27.7379	0	magnitude	total	S0
22	0.846154	0.999069	0.986342	0.0305122	7.78786	30.3105	0	magnitude	total	S0
24	0.923077	0.999567	0.993648	0.020808	5.31097	33.6354	0	magnitude	total	S0

n_co eff	coeff_ratio	eta_total	eta_ac	eps_l2	rmse	snr_db	max_imag	strategy	basis	id
26	1	1	1	2.50703e-15	6.39889e-13	292.017	0	magnitude	total	S0
2	0.0769231	0.032907	0.482746	0.719204	47.9271	2.86296	0	magnitude	ac	S0
4	0.153846	0.0474813	0.696551	0.550862	36.709	5.17914	0	magnitude	ac	S0
6	0.230769	0.0549544	0.806181	0.440249	29.3378	7.12604	0	magnitude	ac	S0
8	0.307692	0.0588038	0.862652	0.370605	24.6968	8.62177	0	magnitude	ac	S0
10	0.384615	0.0608299	0.892374	0.328064	21.8619	9.68082	0	magnitude	ac	S0
12	0.461538	0.0621278	0.911414	0.297634	19.834	10.5264	0	magnitude	ac	S0
13	0.5	0.0632469	0.927831	0.268642	17.9021	11.4165	0	magnitude	ac	S0
15	0.576923	0.0643565	0.94411	0.236411	15.7543	12.5266	0	magnitude	ac	S0
17	0.653846	0.0654477	0.960118	0.199705	13.3082	13.9922	0	magnitude	ac	S0
19	0.730769	0.0664829	0.975303	0.157152	10.4725	16.0736	0	magnitude	ac	S0
21	0.807692	0.0672353	0.986342	0.116866	7.78786	18.6462	0	magnitude	ac	S0
23	0.884615	0.0677334	0.993648	0.0796975	5.31097	21.9711	0	magnitude	ac	S0
25	0.961538	0.0681663	1	9.47025e-15	6.31089e-13	280.473	0	magnitude	ac	S0
1	0.0384615	1.48148e-33	0	1	0.707107	0	0	lowpass	total	S1
3	0.115385	2.12389e-31	2.10908e-31	1	0.707107	-4.82164e-16	0	lowpass	total	S1
5	0.192308	1.86011e-30	1.85863e-30	1	0.707107	-9.64327e-16	0	lowpass	total	S1
7	0.269231	1	1	3.10088e-15	2.19265e-15	290.17	0	lowpass	total	S1

n_coeff	coeff_ratio	eta_total	eta_ac	eps_l2	rmse	snr_db	max_imag	strategy	basis	id
9	0.346154	1	1	3.16873e-15	2.24063e-15	289.982	0	lowpass	total	S1
11	0.423077	1	1	3.12875e-15	2.21236e-15	290.093	0	lowpass	total	S1

First 60 of 324 rows; the complete table is in `tables/table07_selection_curves.csv`.

E3. Energy thresholds, all corpus members

All 96 threshold measurements. The comparison between the two bases for any single signal is the practical statement of the confound.

id	basis	threshold	achieved	n_coeff	coeff_ratio	eps_l2	rmse	snr_db
S0	total	0.5	0.931834	1	0.0384615	0.261087	66.6391	11.6643
S0	total	0.75	0.931834	1	0.0384615	0.261087	66.6391	11.6643
S0	total	0.8	0.931834	1	0.0384615	0.261087	66.6391	11.6643
S0	total	0.9	0.931834	1	0.0384615	0.261087	66.6391	11.6643
S0	total	0.95	0.964741	3	0.115385	0.187775	47.9271	14.5273
S0	total	0.99	0.990637	9	0.346154	0.0967602	24.6968	20.2861
S0	total	0.995	0.995081	14	0.538462	0.0701389	17.9021	23.0808
S0	total	0.999	0.999069	22	0.846154	0.0305122	7.78786	30.3105
S0	ac	0.5	0.696551	4	0.153846	0.550862	36.709	5.17914
S0	ac	0.75	0.806181	6	0.230769	0.440249	29.3378	7.12604
S0	ac	0.8	0.806181	6	0.230769	0.440249	29.3378	7.12604
S0	ac	0.9	0.911414	12	0.461538	0.297634	19.834	10.5264
S0	ac	0.95	0.960118	17	0.653846	0.199705	13.3082	13.9922

id	basis	threshold	achieved	n_coeff	coeff_ratio	eps_l2	rmse	snr_db
S0	ac	0.99	0.993648	23	0.884615	0.0796975	5.31097	21.9711
S0	ac	0.995	1	25	0.961538	9.47025e-15	6.31089e-13	280.473
S0	ac	0.999	1	25	0.961538	9.47025e-15	6.31089e-13	280.473
S1	total	0.5	1	2	0.0769231	3.00469e-15	2.12463e-15	290.444
S1	total	0.75	1	2	0.0769231	3.00469e-15	2.12463e-15	290.444
S1	total	0.8	1	2	0.0769231	3.00469e-15	2.12463e-15	290.444
S1	total	0.9	1	2	0.0769231	3.00469e-15	2.12463e-15	290.444
S1	total	0.95	1	2	0.0769231	3.00469e-15	2.12463e-15	290.444
S1	total	0.99	1	2	0.0769231	3.00469e-15	2.12463e-15	290.444
S1	total	0.995	1	2	0.0769231	3.00469e-15	2.12463e-15	290.444
S1	total	0.999	1	2	0.0769231	3.00469e-15	2.12463e-15	290.444
S1	ac	0.5	1	2	0.0769231	3.00847e-15	2.12731e-15	290.433
S1	ac	0.75	1	2	0.0769231	3.00847e-15	2.12731e-15	290.433
S1	ac	0.8	1	2	0.0769231	3.00847e-15	2.12731e-15	290.433
S1	ac	0.9	1	2	0.0769231	3.00847e-15	2.12731e-15	290.433
S1	ac	0.95	1	2	0.0769231	3.00847e-15	2.12731e-15	290.433
S1	ac	0.99	1	2	0.0769231	3.00847e-15	2.12731e-15	290.433
S1	ac	0.995	1	2	0.0769231	3.00847e-15	2.12731e-15	290.433
S1	ac	0.999	1	2	0.0769231	3.00847e-15	2.12731e-15	290.433
S2	total	0.5	0.814638	4	0.153846	0.430537	0.303985	7.31979
S2	total	0.75	0.814638	4	0.153846	0.430537	0.303985	7.31979

id	basis	threshold	achieved	n_coeff	coeff_ratio	eps_l2	rmse	snr_db
S2	total	0.8	0.814638	4	0.153846	0.430537	0.303985	7.31979
S2	total	0.9	0.906946	8	0.307692	0.305047	0.215382	10.3127
S2	total	0.95	0.959149	14	0.538462	0.202117	0.142707	13.8879
S2	total	0.99	0.990971	22	0.846154	0.095023	0.0670919	20.4434
S2	total	0.995	0.997033	24	0.923077	0.0544735	0.0384615	25.2763
S2	total	0.999	1	25	0.961538	2.95471e-15	2.08621e-15	290.59
S2	ac	0.5	0.814638	4	0.153846	0.430537	0.303985	7.31979
S2	ac	0.75	0.814638	4	0.153846	0.430537	0.303985	7.31979
S2	ac	0.8	0.814638	4	0.153846	0.430537	0.303985	7.31979
S2	ac	0.9	0.906946	8	0.307692	0.305047	0.215382	10.3127
S2	ac	0.95	0.959149	14	0.538462	0.202117	0.142707	13.8879
S2	ac	0.99	0.990971	22	0.846154	0.095023	0.0670919	20.4434
S2	ac	0.995	0.997033	24	0.923077	0.0544735	0.0384615	25.2763
S2	ac	0.999	1	25	0.961538	2.95447e-15	2.08603e-15	290.59
S3	total	0.5	0.941176	1	0.0384615	0.242536	0.851469	12.3045
S3	total	0.75	0.941176	1	0.0384615	0.242536	0.851469	12.3045
S3	total	0.8	0.941176	1	0.0384615	0.242536	0.851469	12.3045
S3	total	0.9	0.941176	1	0.0384615	0.242536	0.851469	12.3045
S3	total	0.95	0.981744	3	0.115385	0.135113	0.474342	17.386
S3	total	0.99	0.996349	5	0.192308	0.0604245	0.212132	24.3757
S3	total	0.995	0.996349	5	0.192308	0.0604245	0.212132	24.3757

id	basis	threshold	achieved	n_coeff	coeff_ratio	eps_l2	rmse	snr_db
S3	total	0.999	1	7	0.269231	1.09948e-15	3.85995e-15	299.176
S3	ac	0.5	0.689655	2	0.0769231	0.557086	0.474342	5.08155
S3	ac	0.75	0.937931	4	0.153846	0.249136	0.212132	12.0713
S3	ac	0.8	0.937931	4	0.153846	0.249136	0.212132	12.0713
S3	ac	0.9	0.937931	4	0.153846	0.249136	0.212132	12.0713
S3	ac	0.95	1	6	0.230769	4.55861e-15	3.88152e-15	286.823
S3	ac	0.99	1	6	0.230769	4.55861e-15	3.88152e-15	286.823
S3	ac	0.995	1	6	0.230769	4.55861e-15	3.88152e-15	286.823
S3	ac	0.999	1	6	0.230769	4.55861e-15	3.88152e-15	286.823
S4	total	0.5	0.573911	6	0.230769	0.652755	0.71847	3.705
S4	total	0.75	0.791897	12	0.461538	0.456183	0.502109	6.81721
S4	total	0.8	0.849561	14	0.538462	0.387864	0.426912	8.22641
S4	total	0.9	0.928473	18	0.692308	0.267445	0.29437	11.4553
S4	total	0.95	0.957847	20	0.769231	0.205312	0.225982	13.7517
S4	total	0.99	0.997312	23	0.884615	0.0518493	0.0570691	25.7051
S4	total	0.995	0.997312	23	0.884615	0.0518493	0.0570691	25.7051
S4	total	0.999	1	25	0.961538	3.58952e-15	3.95089e-15	288.899
S4	ac	0.5	0.573911	6	0.230769	0.652755	0.71847	3.705
S4	ac	0.75	0.791897	12	0.461538	0.456183	0.502109	6.81721
S4	ac	0.8	0.849561	14	0.538462	0.387864	0.426912	8.22641
S4	ac	0.9	0.928473	18	0.692308	0.267445	0.29437	11.4553

id	basis	threshold	achieved	n_coeff	coeff_ratio	eps_l2	rmse	snr_db
S4	ac	0.95	0.957847	20	0.769231	0.205312	0.225982	13.7517
S4	ac	0.99	0.997312	23	0.884615	0.0518493	0.0570691	25.7051
S4	ac	0.995	0.997312	23	0.884615	0.0518493	0.0570691	25.7051
S4	ac	0.999	1	25	0.961538	3.58819e-15	3.94943e-15	288.902
S5	total	0.5	0.651755	6	0.230769	0.590123	0.145738	4.58116
S5	total	0.75	0.797136	10	0.384615	0.450404	0.111233	6.92796
S5	total	0.8	0.845012	12	0.461538	0.393685	0.0972253	8.09701
S5	total	0.9	0.912491	16	0.615385	0.29582	0.0730562	10.5795
S5	total	0.95	0.957423	20	0.769231	0.206343	0.0509588	13.7082
S5	total	0.99	0.991894	24	0.923077	0.090032	0.0222345	20.9121
S5	total	0.995	1	25	0.961538	2.04521e-15	5.05089e-16	293.785
S5	total	0.999	1	25	0.961538	2.04521e-15	5.05089e-16	293.785
S5	ac	0.5	0.651755	6	0.230769	0.590123	0.145738	4.58116
S5	ac	0.75	0.797136	10	0.384615	0.450404	0.111233	6.92796
S5	ac	0.8	0.845012	12	0.461538	0.393685	0.0972253	8.09701
S5	ac	0.9	0.912491	16	0.615385	0.29582	0.0730562	10.5795
S5	ac	0.95	0.957423	20	0.769231	0.206343	0.0509588	13.7082
S5	ac	0.99	0.991894	24	0.923077	0.090032	0.0222345	20.9121
S5	ac	0.995	1	25	0.961538	2.03057e-15	5.01474e-16	293.848
S5	ac	0.999	1	25	0.961538	2.03057e-15	5.01474e-16	293.848